

Subnetting, ARP, DHCP, NAT, and NMAP

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 3 hours and 15 minutes.

MAC Addresses + ARP

Article: [What is a MAC Address and How Do I Find It? | TechTarget](#)

What is a MAC address?

(Media Access Control Address): A 12-digit hexadecimal number assigned to each device connected to the network. It is usually specified as a unique identifier during device manufacturing, and is found on a device's network interface card. It is required when trying to locate a device or when performing diagnostics on a network device. It belongs to the data link layer.

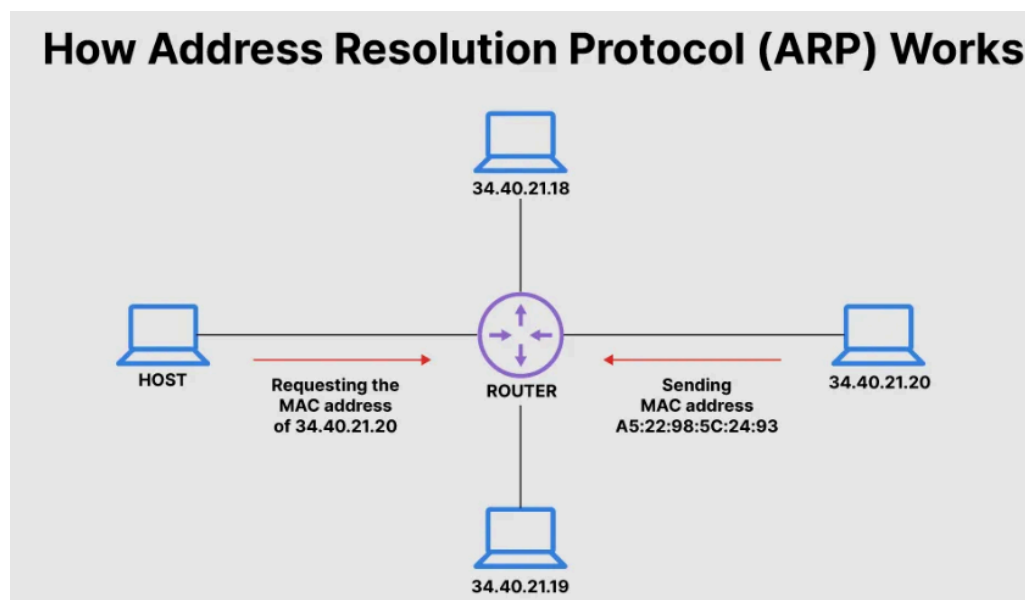
IP addresses are logical network identity. MAC addresses are physical/local hardware identity.

Ipconfig /all will give the MAC address next to **Physical Address**.

Article: [What Is ARP \(Address Resolution Protocol\)? How Does It Work?](#)

What is ARP?

(Address Resolution Protocol): Connects an ever-changing IP address to a MAC address in a LAN.



ARP = “Who has this IP address?”

It maps IP addresses to MAC addresses inside local networks.

The command **arp -a** will list IP addresses, MAC addresses, and ARP cache. Try identifying: your router, local devices, and repeated MAC vendors.

Router: Default Gateway

Why is ARP spoofing dangerous?

Also known as ARP poisoning, and it is a cyberattack where an attacker tricks devices on a local network into sending data to the attacker instead of the real destination. In ARP spoofing, the attacker sends fake ARP replies to devices on the network.

For example: the attacker tells your computer: “Hey, I’m the router. My MAC address is AA:BB:CC...”, but that MAC address actually belongs to the attacker. Now your computer will send traffic meant for the router to the attacker instead.

Therefore, they can intercept traffic, allowing them to read passwords, messages, login session, and unencrypted data.

DHCP + NAT

Article: [Dynamic Host Configuration Protocol \(DHCP\) - GeeksforGeeks](#)

What is DHCP?

(Dynamic Host Configuration Protocol): Seamless network connectivity is achieved by automatically assigning IP addresses and configuration settings to devices using DHCP. It assigns a unique IP address to each device, provides the subnet mask to define the local network range, configures the default gateway for communication outside the local network, and supplies DNS server addresses for domain name resolution.

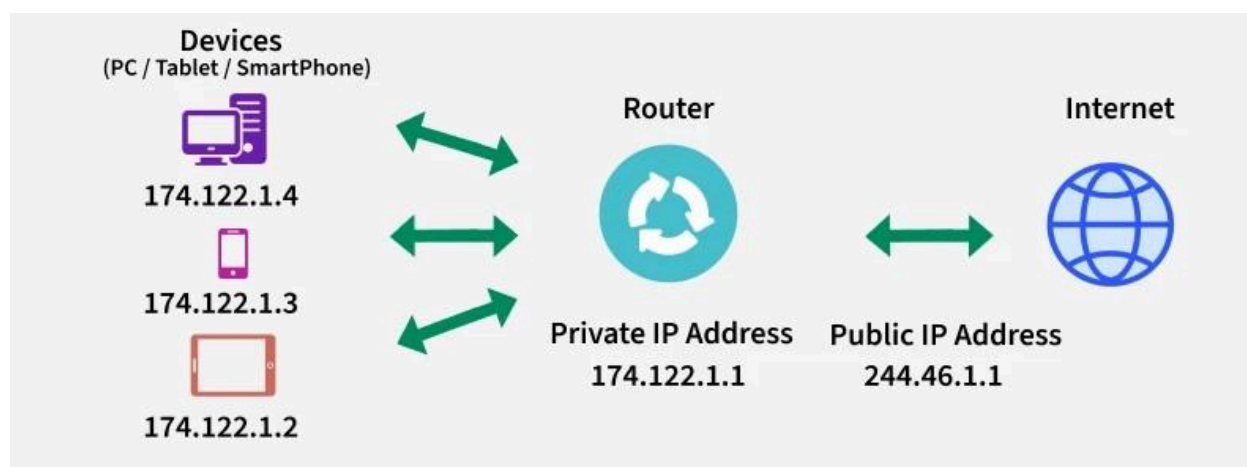
DHCP automates IP address, default gateway, subnet mask, and DNS server for every device.

Run **ipconfig /all** to find DHCP Enabled and DHCP Server. Also, Lease Obtained and Lease Expires.

Article: [Network Address Translation \(NAT\) - GeeksforGeeks](#)

What is NAT?

(Network Address Translation): allows multiple devices in a private network to access the internet using a single public IP address.



What is private vs public IP addressing?

A private IP is used inside local networks like: home Wi-Fi, schools, and businesses. They are not accessible directly from the internet. All your devices like phone and laptop get all private IPs from your router. A public IP address is the address visible on the internet. It is assigned by your ISP (Internet Service Provider).

To find private IP: ipconfig

To find public IP: google "what is my IP"

Subnetting Foundations

Video: [What is Subnetting? - Subnetting Mastery - Part 1 of 7](#)

What is subnetting?

Taking a network and dividing it into sub-networks. The seven attributes of subnetting are: Network ID, Broadcast IP, First Host IP, Last Host IP, Next Network, # IP Addresses, and CIDR/Subnet

Subnet Mask: Separates Network / host

CIDR: slash notation (/24)

Network Address: identifies network (**192.1.168.1**)

Host Address: identifies device (192.1.168.1)

How many hosts can a /24 network support?

IPv4 are 32 bits total. 254 usable hosts. /24 means 24 bits for the network, so 8 left for host. $2^8 - 2$. Because 1 address = network address, and 1 address = broadcast address.

192.168.1.0 = network address

192.168.1.255 = broadcast address

What does /16 vs /24 mean?

16 network bits, so 16 host bits. $2^{16} - 2$. 65534 usable hosts.

Mini Task: 192.168.1.25/24

Network address: 192.168.1.0

Likely gateway: 192.168.1.1

Host range: 192.168.1.1 to 192.168.1.254

NMAP Intro

Article: [Chapter 15. Nmap Reference Guide | Nmap Network Scanning](#)

What is NMAP used for?

Open source tool for network exploration and security auditing. It was designed to rapidly scan large networks. It uses raw IP packets to determine what hosts are available on the network, what services those hosts are offering, what operating systems they are running, what type of packet filters/firewalls are in use, and more.

Run **nmap localhost**. This shows open ports, services, and states.

Ports: 135/tcp, 445/tcp, 9010/tcp

States: open

Service: msrpc, microsoft-ds, sdr

What does open vs filtered vs closed mean in NMAP?

Open port: a service is actively listening on that port

Closed port: no application is listening there, but device is reachable since nmap got a response back

Filtered: nmap cannot determine whether the port is open because packets are being blocked or dropped

Now **ipconfig** to find IP then scan it with nmap. **Nmap your_ip**

Also **nmap -sV localhost**

-sV tells nmap to determine what service is running on an open port and try to identify its version

What is service fingerprinting?

The process of identifying what service is running on a port, what software it is, and sometimes the exact version and operating system.

Try **nmap -A localhost**

Aggressive scan, automatically performs OS detection, version detection (-sV), scripting scanning (-sC), and traceroute

Script Scanning: run small scripts against a target to gather more information or test for vulnerabilities

- Identify services, retrieve webpage titles, enumerate users/shares, detect vulnerabilities, gather SSL info, brute-force weak credentials

Next, find local subnet. To do this get Subnet Mask and IPv4 address. **Nmap -sn my_subnet**
This will discover live hosts. -sn flag means: ping scan, no port scanning.